Paleomagnetism of the Triassic Guichon Batholith and Rotation in the Interior Plateau, British Columbia

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The Guichon Batholith, located near the south end of the Interior Plateau in south-central British Columbia, is composed of unmetamorphosed massive felsic intrusive rocks in several distinct phases (Northcote 1969). Stratigraphic and radiometric evidence indicate that the batholith was emplaced during the Late Triassic (198 ± 8 m.y.) and unroofed by Early Jurassic. Analysis of the remanence of 92 cores (184 specimens) from 19 representative sites led to the isolation of a stable primary remanent magnetism at 15 sites after alternating-field demagnetization. Variance ratio analysis of the remanence directions indicates that the phases cannot be distinguished by the paleomagnetic method. This supports the evidence from contact relationships and K–Ar isotopic dating of biotites that the phases cooled nearly contemporaneously. The pole position determined for the Guichon Batholith (12.9° E, 65.6° N) is discordant with other Upper Triassic pole positions determined for North American formations. The discordance may be explained by a clockwise rotation 40° ± 10° of the batholith and surrounding rocks in the southern end of the Interior Plateau, with most of the Plateau to the north acting as a stable non-rotated tectonic block. Other evidence is cited which is consistent with this hypothesis.

Le batholite de Guichon, près de la terminaison sud du plateau intérieur de la Colombie Britannique du Sud, est formé de divers faciès de roches intrusives felsiques non métamorphiques (Northcote 1969). La stratigraphie et la géochronologie indiquent une mise en place tardi-triassique (198 ± 8 m.a.) et une érosion éojurassique. Les analyses du magnétisme rémanant de 92 carottes (184 échantillons) provenant de 19 lieux choisis permit de mettre en évidence un magnétisme rémanent primaire à 15 localités, après démagnétisation par champ alternatif. L'analyse statistique du magnétisme rémanant montre que l'on ne peut distinguer les phases d'après le paléomagnétisme. Ceci est en accord avec les observations des contacts et avec les dates K–Ar sur les biotites, qui montrent que les phases se sont refroidies en même temps. La position du pôle déterminée pour le batholite de Guichon (12.9° E, 65.6° N) est en désaccord avec d'autres pôles du Trias supérieur, déterminés sur des formations nord-américaines. Cette anomalie peut s'expliquer par une rotation de 40° ± 10° dans le sens des aiguilles d'une montre impliquant le batholite et son encaissant dans le Sud du Plateau intérieur, la plus grande partie du plateau étant un bloc tectonique stable. Des arguments complémentaires viennent étayer cette hypothèse.

Introduction

The Guichon Batholith is a semi-concordant granodiorite pluton about 40 miles (64 km) long by 17 miles (27 km) wide (Fig. 1) located near the southern end of the Interior Plateau in south-central British Columbia. The geology of the batholith has been mapped on a regional scale by Cockfield (1948) and Duffell and McTaggart (1952). Since then several areas of the batholith including the Craigmont and Bethlehem copper deposits have been mapped in more detail (White *et al.* 1957; Carr 1960, 1961, 1963, 1966; Rennie *et al.* 1961) and dated radiometrically (Baadsgaard *et al.* 1961; Wanless *et al.* 1965, 1968; Dirom 1965; White *et al.* 1967; Chrismas *et al.* 1969). Recently Northcote (1969) published the results of detailed geologic mapping, petrologic investigations, and K–Ar dating of the various intrusive phases of the entire batholith. This paleomagnetic study was initiated to determine if the phases of the batholith could be distinguished on the basis of remanence direction, and if the batholith has been subjected to regional tectonic rotation.

Geology

Petrology

Based on mineral proportions, textural differences, and intrusive relationships, Northcote (1969) subdivided the Guichon Batholith into the: (1) Hybrid quartz diorite phase; (2) Highland Valley phase including the Guichon quartz

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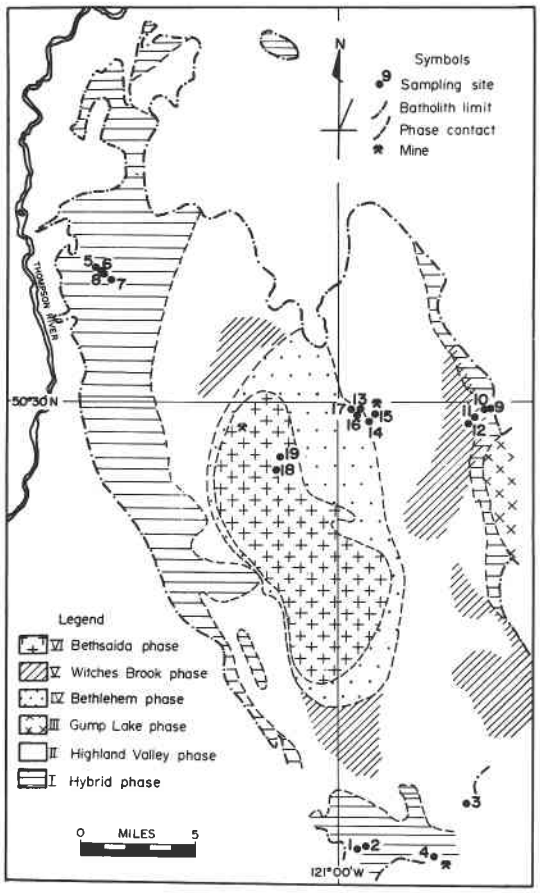


FIG. 1. Generalized geology after Northcote (1969) and sample sites in the Guichon Batholith.

diorite variety grading into the Chataway quartz diorite–granodiorite variety containing irregular dike-like bodies of the LeRoy granodiorite variety; (3) Gump Lake coarse-grained or porphyritic granodiorite–quartz monzonite phase; (4) Bethlehem granodiorite phase; (5) Whitches Brook mottled pink and gray granodiorite phase found as irregular dike-like bodies in phases (1) to (4); (6) Bethsaida granodiorite–quartz monzonite phase; and (7) Gnawed Mountain porphyry and other leucocratic rocks found as irregular dike-like bodies in all older phases. The major phases are concentrically arranged (Fig. 1) progressing from the older marginal mesozonal phases (1) and (2) through phases of intermediate age (3), (4), and (5) to the younger epizonal phases (6) and (7) of the core. In general, the mean percentages of magnetite and mafic minerals and the mean specific gravity decreases progressively from the oldest (1) to youngest (7) phase while the mean percentages of quartz and orthoclase tend to increase. The lack of chilled contacts between phases indicates that they cooled nearly simultaneously. Preservation of the original textures and structures indicates that the batholith has not been metamorphosed since emplacement. Foliation decreases in intensity from moderate in the marginal phases to slight in the core phases. In general, the foliation strikes parallel to and dips toward the margins of the batholith. The sulfide mineralization is found in altered, sheared, fault zones which strike north–south and dip steeply. The nature and magnitude of the fault displacements is not known.

Geochronology

The Guichon Batholith intrudes sedimentary and volcanic rocks of the Nicola Group of Upper Triassic age (Karnian and Norian—Frebold and Tipper 1969) and older rocks. The batholith is unconformably overlain by Lower Jurassic sediments (upper Hettangian and Sinemurian—Frebold and Tipper 1969) containing clasts from the batholith and by younger rocks. Stratigraphically, therefore, the batholith was emplaced during the Late Triassic (upper Norian or Rhaetian) and/or Lower Jurassic (lower Hettangian) and it was immediately unroofed. The intruded Nicola Group has been contact metamorphosed to hornfels, but it does not appear to have been regionally metamorphosed.

Baadsgaard *et al.* (1961) determined a radiometric age of 186 ± 9 m.y. for the batholith using the K–Ar method on biotite. Dirom (1965) (in White *et al.* 1967) obtained a concordant average age of 200 ± 8 m.y. on eight samples from the batholith using the K–Ar method on biotite and hornblende excluding one porphyry sample which gave 171 ± 12 m.y. for which a second sample gave 199 ± 8 m.y. White *et al.* (1967) reported 26 K–Ar dates on biotites from all 7 intrusive phases of the batholith. They obtained an average of 198 ± 8 m.y. and were not able to distinguish between the phase ages. Wanless *et al.* (1968) gave 4 K–Ar ages on biotite and hornblende with an average of 189 ± 9 m.y. Finally, Chrismas *et al.* (1969) obtained a Rb–Sr whole-rock isochron age of 200 ± 2 m.y. for batholith rocks in the Craigmont Mine ore zone, and on the basis of this result and sulfur, carbon,

and oxygen isotope results concluded that both host and ore formed by the rapid cooling and differentiation of a basic magma with the ore representing the terminal cooling stage.

In the light of present knowledge, therefore, it appears that the Guichon Batholith was emplaced during the Late Triassic about 198 ± 8 m.y. ago; all intrusive phases cooled essentially contemporaneously; the batholith was unroofed and being eroded by Early Jurassic; and, it has not been metamorphosed since emplacement.

Sampling and Measurement

Ninety-two (92) cores, oriented *in situ* by sun compass, were collected from 19 sites on the batholith (Fig. 1) with the majority of sites representing the Hybrid (1) and Highland Valley (2) phases because of their higher magnetite content. Two specimens were cut from each core and their natural remanent magnetizations (n.r.m.) measured using methods described by Larochelle and Christie (1967). One pilot specimen from each site was demagnetized (Larochelle and Black 1965) stepwise in alternating field (a.f.) peak intensities of 50, 100, 200, 300, 400, 500, 650, and 800 Oe with its remanence remeasured after each step. Guided by the stability index (s.i.) data calculated for the pilot specimens (Tarling and Symons 1967), the remaining specimens from each site were a.f. demagnetized at a selected field intensity and their remanent magnetization remeasured. The specimens from 11 sites with generally low s.i. values were demagnetized in a second higher field intensity and remeasured.

Analysis

The presence of unstable remanence components is apparent from the s.i. values as 9 of the 19 pilot specimens give values below 2.5. Most specimens appear to acquire anhysteretic remanent magnetization (a.r.m.) during a.f. demagnetization above the 300 to 500 Oe step (Doell and Cox 1967) because all pilot specimens show increased scatter in remanence directions and 11 of the 19 pilot specimens show irregular variations in remanence intensity from step to step. Experiments in progress by K. W. Christie (personal communication) confirm this interpretation for several of the specimens. Anhyster-

TABLE 1. Sampling and remanence data by sites

	Location					Cores				Site mean remanence			
Site	° W	° N	Phase	s.i.	*H*	*N*	*a*	*b*	*n*	*D*	*I*	*R*	s.e.
1	120.98	50.20	I	1.1	100, (800)	5	2	1	2	15.3	63.9	1.9991	1.7
2	120.97	50.19	I	0.7	100, (800)	4	4		0				
3	120.88	50.24	II	1.5	100, (800)	5			5	16.1	62.1	4.8878	6.1
4	120.90	50.21	I	7.0	300, (650)	5	1		4	26.5	71.4	3.4717	17.0
5	121.23	50.58	I	5.0	200	5	2		3	46.2	48.5	2.9132	9.8
6	121.23	50.58	I	7.9	200, (650)	5		1	4	27.7	25.6	3.9671	4.3
7	121.22	50.58	I	1.2	300	5			5	25.8	43.2	4.8494	7.0
8	121.23	50.58	I	8.0	300	5			5	33.5	34.1	4.8908	6.0
9	120.86	50.49	I	3.1	300, (650)	5	1		4	27.6	35.2	3.8828	8.0
10	120.86	50.49	II	2.5	200, (500)	5	2		3	15.1	62.5	2.5756	21.5
11	120.87	50.48	II	2.4	200	5			5	359.5	53.8	4.9539	3.9
12	120.87	50.48	V	3.2	300, (650)	5	2		3	20.1	61.7	2.8358	13.4
13	120.98	50.50	II	5.7	400	5	1		4	196.6	−16.3	3.9847	2.9
14	120.97	50.49	VII	7.3	200	5			5	10.2	58.6	4.9016	5.7
15	120.96	50.49	II	2.4	200	5	1		4	23.9	60.6	3.9802	3.3
16	120.98	50.49	VII	9.2	100, (800)	5		1	4	7.3	73.8	3.9815	3.2
17	120.99	50.50	IV	29.0	300	4			4	209.7	−30.6	3.9569	4.9
18	121.06	50.46	VI	2.7	100, (800)	5	3		2	218.1	75.0	1.9819	7.7
19	121.05	50.46	VI	1.0	200, (650)	4	1	1	2	352.3	41.7	1.9961	3.5

NOTES—° W, ° N: longitude west and latitude north of site in degrees. Phase sampled: I—Hybrid; II—Highland Valley; IV—Bethlehem; V—Witches Brook; VI—Bethsaida; and VII—Gnawed Mountain and other porphyries. s.i.: Stability index (Tarling and Symons 1967) of pilot specimen of a.f. demagnetization. *H*: a.f. demagnetization field peak intensity in oersteds with brackets about a second intensity attempted but not used in the statistics. Cores: collected (N), rejected because specimen remanence directions diverged by $>20°$ (*a*) or because anomalously directed compared to remaining cores at site (*b*), and used (*n*) for site remanence statistics. *D*, *I*, s.e.: declination, inclination, and standard error of site mean remanence direction in degrees. *R*: length of resultant giving *n* cores unit weight. The Fisher statistics can be found using: $k = (n-1)(n-R)$, and $\alpha_{95} = 140/(kn)^{\frac{1}{2}}$.

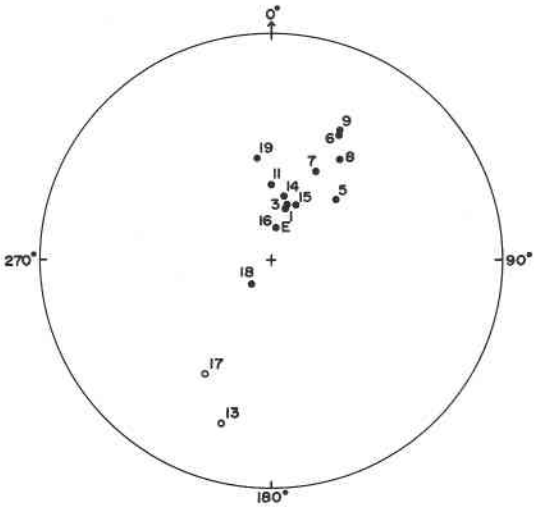


FIG. 2. Equal-area stereographic projection showing site mean remanence directions with positive (negative) or downwardly (upwardly) inclinations indicated by solid (open) circles and the earth's present magnetic field direction (E).

etic remanent magnetization is also the cause of the increased scatter in remanence direction found in all sites upon a.f. demagnetization in the second higher field. Accordingly, the results of the second "cleaning" are not considered further.

In 20 cores the angle θ between the remanence directions of the two specimens exceeds 20°, and therefore these cores are considered inhomogenously magnetized and are rejected from further consideration (Table 1). In 4 sites the remanence direction of one core deviated by more than 3 times the standard deviation of the remaining cores from their mean direction and so these 4 anomalous cores are also rejected. After calculating the site mean remanence directions and their standard errors (s.e.), the s.e. in sites 4, 10, and 12 exceeds 10°. These 3 sites are considered to have insufficiently homogenous remanence magnetizations and are rejected. Thus, 15 of the 19 sites (79 %) are considered to have reliable homogenous remanence directions (Fig. 2).

Discussion

Remanence Stability

Several features indicate that a primary stable remanence of probable thermoremanent or high-temperature chemical remanent origin is isolated in the 15 reliable sites. First, the removal of unstable viscous remanence (v.r.m.) components is shown by: (*a*) the marked 50 to 80 % decrease in n.r.m. intensity in 17 of 19 pilot specimens produced by the 50 Oe cleaning combined with decreases of only 10 to 30 % thereafter; (*b*) the initial movement in remanence direction produced by the 50 Oe cleaning is away from the earth's present magnetic field direction in nearly all pilot specimens; and (*c*) the earth's present magnetic field direction lies within the cone of 95 % confidence of only one site (16) mean remanence direction. Second, two sites (13 and 17) have reversed remanence directions approximately antiparallel to the remaining 13 reliable site directions (Fig. 2). Third, the petrological studies of Northcote (1969) indicate that magnetite is the sole likely remanence carrier, and that no thermal metamorphic events have transpired since cooling of the batholith to produce a secondary remanence. Finally, there is no evidence such as anomalously intense remanence to suggest the presence of significant lightning-produced i.r.m. in any of the 15 reliable sites.

Angular Variance

The variance ratio test (Larochelle 1969) shows that the ratio of the between-site to within-site variance is significantly greater than the corresponding value of the theoretical statistic F at the 95 % confidence level (i.e. $\delta_b^2/\delta_w^2 > F_{.05}$) which indicates that the site mean magnetic remanence directions are significantly distinct within every phase, within the combined younger phases (4, 6, 7), and within the batholith as a whole (Table 2). Two explanations for this test result are tenable. First, sufficient time elapsed between cooling of the various sites through their Curie Points for secular variation to occur in the earth's ambient paleofield. Time differences sufficient to allow for reversal of the paleofield are indicated as both normal and reversed remanence directions are found in the Highland Valley and Bethlehem phases. Second, and likely less influential because significant faulting is not reported within the batholith, tectonic rotation between the sites may have occurred.

Comparison of the between-phase variance (Table 3) gives a ratio significantly less than the statistic $F_{.05}$ for the Hybrid to Highland Valley phase comparison and for each of these phases when compared to the combined younger phases. Thus, the phase mean magnetic remanence direc-

TABLE 2. Angular variance ratio analysis within phases

Phase†	n	B	D deg	I deg	R	s.e. deg	δ_w^2 radians^2	δ_b^2 radians^2	δ_b^2/δ_w^2	$F_{.05}$
I Hybrid	23	6	30.0	42.0	5.8305	6.1	.0585	.1905	3.25	2.12
II Highland Valley	18	4	13.9	48.9	3.7735	11.2	.0276	.6299	22.80	2.44
IV Bethlehem	4	1	29.7	30.6						
VI Bethsaida	4	2	334.1	69.9	1.7366	29.4	.0220	1.0537	47.96	6.94
VII Gnawed Mountain	9	2	9.2	66.2	1.9824	7.6	.0334	.1566	4.69	3.74
Younger*	17	5	8.1	62.1	4.4742	13.2	.0303	.7095	23.39	2.36
Combined‡	58	15	20.4	50.6	13.8417	6.0	.0406	.5245	12.92	1.62

NOTES: n, number of core mean directions. B, number of sites. D, I, mean declination and inclination of B site mean directions. R, resultant of B unit vectors. s.e., standard error of phase mean. δ_w^2, δ_b^2, within-site and between-site angular variances. $F_{.05}$ theoretical value of $F_{2(B-1)\,2(n-B),\,.05}$.

*Younger includes Bethlehem (IV), Bethsaida (VI), and Gnawed Mountain (VII) phases.

†Polarity of data from sites 13 and 17 reversed.

‡The Fisher statistics for the site mean direction are: $k = 12.1$; and the cone of 95 % confidence, $\alpha_{95} = 10.4°$.

TABLE 3. Angular variance ratio analysis between phases

Phase pair†	B	δ_w^2	δ_b^2	δ_b^2/δ_w^2	$F_{.05}$
Hybrid – Highland Valley	10	.0991	.1269	1.28	3.63
Hybrid – Younger*	11	.1546	.4755	3.08	3.55
Highland Valley – Younger*	9	.2149	.1254	.58	3.74

NOTES: B, number of site means. δ_w^2, δ_b^2, within-phase and between-phase angular variances. $F_{.05}$, theoretical value of $F_{2,\,2(B-2),\,.05}$

*Younger includes Bethlehem (IV), Bethsaida (VI), and Gnawed Mountain Porphyry VII) Phases.

†Polarity of data from sites 13 and 17 reversed.

tions are not significantly different leading to three conclusions: (1) it is unlikely that the various phases of the batholith can be classified by remanence direction, (2) the phases probably cooled simultaneously with the site directional differences reflecting the movement of isotherms during cooling, and (3) the site mean remanence directions should be considered further only as a combined group.

Pole Position

Reversing the polarity of sites 13 and 17 and giving each of the 15 site mean magnetic remanence directions unit weight, the pole position calculated for the Guichon Batholith using the dipole formula is 12.9° E, 65.6° N with semi-axes of 9.4° and 14.0° for its oval of 95 % confidence. This position is clearly discordant when compared to other Upper Triassic pole positions determined for North American formations (Table 4) which cluster about a mean position of about 98° E, 66° N (Fig. 3). The discordance cannot be explained by stable secondary remagnetization or v.r.m. overprinting because, in addition to evidence given above for remanence stability, the Guichon pole position is even more remote from reliable post-Triassic North American pole positions (Symons 1971; Irving 1964) and the earth's present magnetic pole than the



FIG. 3. Northern hemisphere north of latitude 40° N showing the sampling site (S), Guichon pole position (G), and mean Upper Triassic pole position (T) surrounded by reliable Upper Triassic poles (solid circles) from North American formations (Table 4).

mean Upper Triassic position. The discordance is best explained in terms of post-emplacement translation or rotation affecting the batholith as a whole. Translation would require some 35° of total arc movement which is totally inconsistent with the distribution of pre-Triassic rocks in the Cordillera such as the Upper Permian Cache Creek Group (McGugan *et al.* 1964). However, a simple clockwise rotation about a vertical axis of about 37° would bring the Guichon pole to within 2° of the mean Triassic position (Fig. 3) and there is evidence for such rotation.

Rotation

First, the regional distribution of Permo-Carboniferous to Lower Cretaceous volcanic and sedimentary rocks in the Cordillera is indicative of rotation (Douglas 1969; Bostock 1970) (Fig. 4*a*). These rocks are found mostly in the Interior Plateau and Cascade Mountains. They are bounded on the east by Paleozoic rocks—excluding later intrusives—confined mostly to the Columbia Highlands and Mountains, and on the west by felsic intrusive rocks emplaced during the mid-Cretaceous to Early Tertiary Nevadan Orogeny and confined mostly to the Coast Mountains. Assuming that most of the Interior Plateau has been a stable non-rotated block, then both the trace of its eastern and western margins

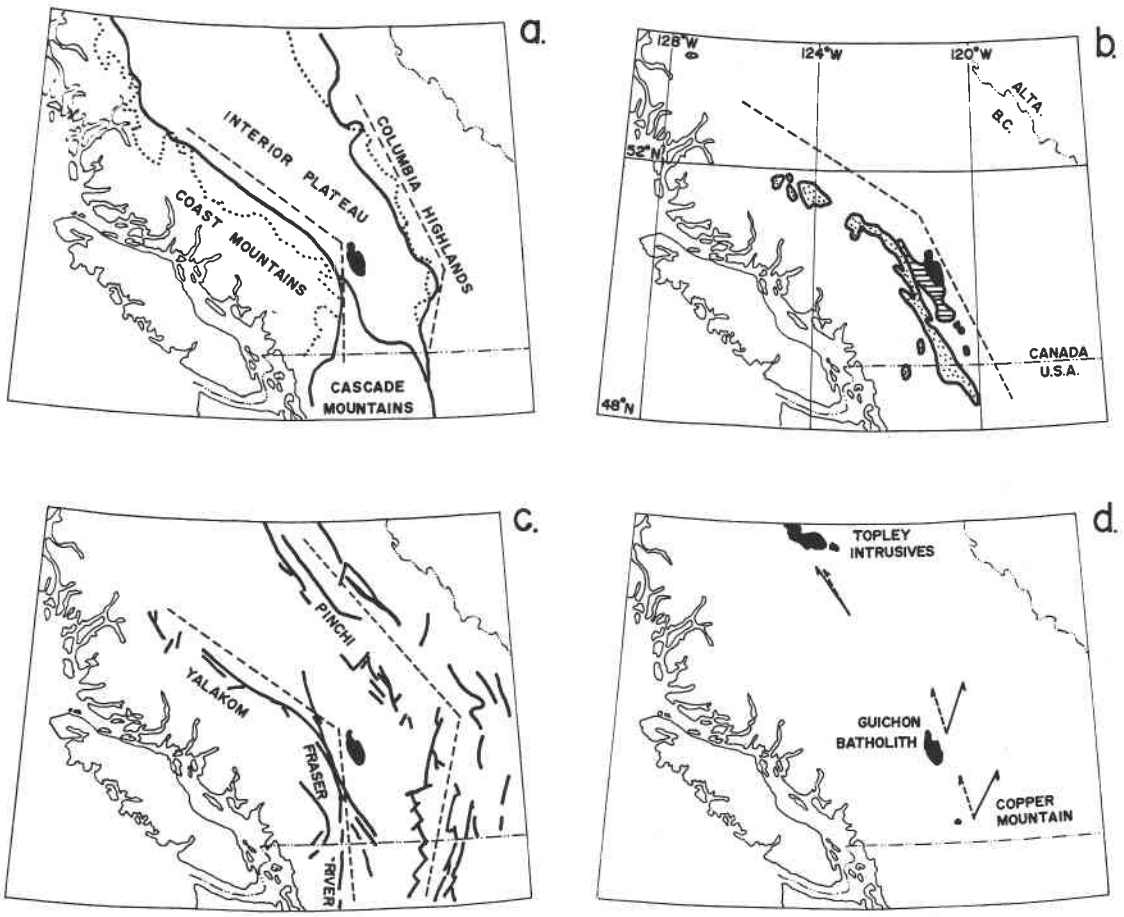


FIG. 4. Sketch map of southern British Columbia showing the Guichon Batholith (solid area) in relation to the: (*a*) Physiographic boundaries by solid lines (Bostock 1970), major lithologic boundaries by dotted lines (Douglas 1969), and their generalized trends by dashed lines; (*b*) Lower Cretaceous sediments (hatched area) with interbedded volcanics (dotted area) (Rudkin 1964) and their generalized trend (dashed line); (*c*) Major faults by solid lines (Stockwell 1969) and their generalized trends by dashed lines; and, (*d*) Mean remanence declinations with the measured direction shown by a solid-line arrow and the theoretical non-rotated direction by a dashed-line arrow.

TABLE 4. Paleomagnetic pole positions derived from North American Formations of Upper Triassic Age

Rock unit	Locality	N	Pole position	Reference*
1 Chinle formation	Utah and New Mexico	3	93° E, 55° N	Collinson and Runcorn (1960)
2*a* North Mountain basalt	Nova Scotia	17	113° E, 66° N	Larochelle (1967)
2*b* North Mountain basalt	Nova Scotia	25	104° E, 73° N	Carmichael and Palmer (1968)
3 Grand Manan lava	New Brunswick	4	100° E, 80° N	Carmichael and Palmer (1968)
4 Great Dyke	Nova Scotia	2	98° E, 69° N	Larochelle and Wanless (1966)
5 Newark Group	Massachusetts	5	88° E, 55° N	Irving and Banks (1961)
6 Newark Group	Connecticut	53	87° E, 65° N	de Boer (1968)
7 Newark Group	New Jersey	29	108° E, 63° N	Opdyke (1961)
8 Newark Group	Pennsylvania	20	105° E, 62° N	Beck (1965)

NOTES: N, number of site means on which the derived pole position is based.
*More complete information is given in a summary table in Symons (1971).

change clockwise by about 40° and 50° respectively at the southern end of the Plateau where the Guichon Batholith is located.

Second, the change in trend is seen in individual post-Guichon-emplacement pre-Nevadan–Orogeny formations of the Plateau such as the Lower Cretaceous sediments (Rudkin 1964) (Fig. 4*b*). Going from northwest to south, the trend of these units changes in a clockwise sense by about 30°.

Third, the major faults of south-central British Columbia tend to follow the margins of the Plateau (Stockwell 1969) (Fig. 4*c*). On the eastern margin, following the Pinchi–McLeod Lake fault system towards the south, the fault traces rotate clockwise by about 50° on reaching the Vernon and Greenwood map-areas. Similarly on the western margin following the Yalakom–Tchaikazan fault system to the south, the fault traces rotate clockwise by about 50° upon forming the Frazer River – Hozameen fault system. Because of the paucity of comparable units on either side of these fault systems, little is known of the sense and magnitude of their horizontal displacements (Campbell 1961, 1963; Cockfield 1948; Duffell and McTaggart 1952; Jones 1959; Monger 1970; Muller and Tipper 1962; Rice 1960; Tipper 1959, 1960, 1963, 1968; Tipper and Campbell 1966) or the evidence is contradictory (Monger 1968). Tipper (1969) reports a right-lateral movement of 140 miles (225 km) or more on the Yalakom–Tchaikazan fault system in the Mount Waddington map-area. McTaggart (1970) suggests a right-lateral displacement on the Fraser River system from drag folds; and the offsetting of the Permo-Carboniferous sediments, Mesozoic Coquihalla serpentine belt, and the Lower Cretaceous Jackass Mountain Group in the Ashcroft (Duffell and McTaggart 1952) and Hope (Monger 1970) map-areas suggest a displacement of 40 miles (64 km) or more. Such displacements are compatible with rotation of the southern end of the Plateau.

Fourth, a preliminary examination of paleomagnetic results currently under study by the author are consistent with the rotation hypothesis (Fig. 4). The Upper Jurassic phases of the Topley Intrusives near the center of the Interior Plateau give a pole position similar to other North American Jurassic formations indicating that the Intrusives are located in a stable non-rotated block. Conversely, the Late Triassic Copper Mountain and related stocks to the south of the Guichon Batholith give a pole position similar to the Guichon pole thereby also indicating clockwise rotation.

Several other points are favorable to the rotational hypothesis. The center of maximum rotation or hinge is located just to the north of the Guichon Batholith. The interior curve radius on the western side of the Plateau is small compared to the exterior curve radius for the trend changes. The eastern and western margin trends tend to converge to the south as the formations of the Plateau become increasingly deformed until piled up as part of the Cascade Mountains formed during the Nevadan Orogeny (McTaggart 1970). Also, the relative abundance of Mesozoic intrusive and extrusive rocks in the south end of the Interior Plateau when compared to the dominance of Upper Cretaceous–Tertiary extrusives in the north could reflect uplift associated with the rotation during the Nevadan Orogeny. None of the geologic evidence cited need necessarily be explained by rotation, e.g. the sedimentation trends could be explained in terms of primary

basin configuration. However, the paleomagnetic evidence suggests rotation and the geologic evidence can be explained by rotation.

Conclusions

1. The Late Triassic Guichon Batholith retains a stable primary remanent magnetization.

2. It does not appear possible to separate or classify the individual intrusive phases on the basis of their remanence direction which is evidence for their contemporaneous rapid cooling.

3. The Guichon Batholith and surrounding rocks in the southern end of the Interior Plateau were rotated clockwise by about 40° ± 10° during the Nevadan Orogeny while most of the Plateau to the north of the batholith appears to have acted as a stable non-rotated block.

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